

HAT-P-36b

R-1827

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_160403 · created 2026-08-03T05:25:56+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457481.8725 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0896 +/- 0.0093
Transit depth (Rp/Rs)^2	0.8 +/- 0.17 [%]
Orbital Inclination (inc)	89.91 +/- 1.42
Airmass coefficient 1 (a1)	1.148 +/- 0.011
Airmass coefficient 2 (a2)	-0.11 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	None
Optimal Aperture	3.53
Optimal Annulus	10.03
Transit Duration (day)	0.0942 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0896 $\pm$ 0.0093 vs 0.126 (deviation 2.14 σ)
Duration fit vs published	0.0942 d vs 0.0925 d (deviation 0.85 σ)
Mid-time O-C	2.8 min
Depth SNR	5.3
Residual scatter	1.0 %

Light curve

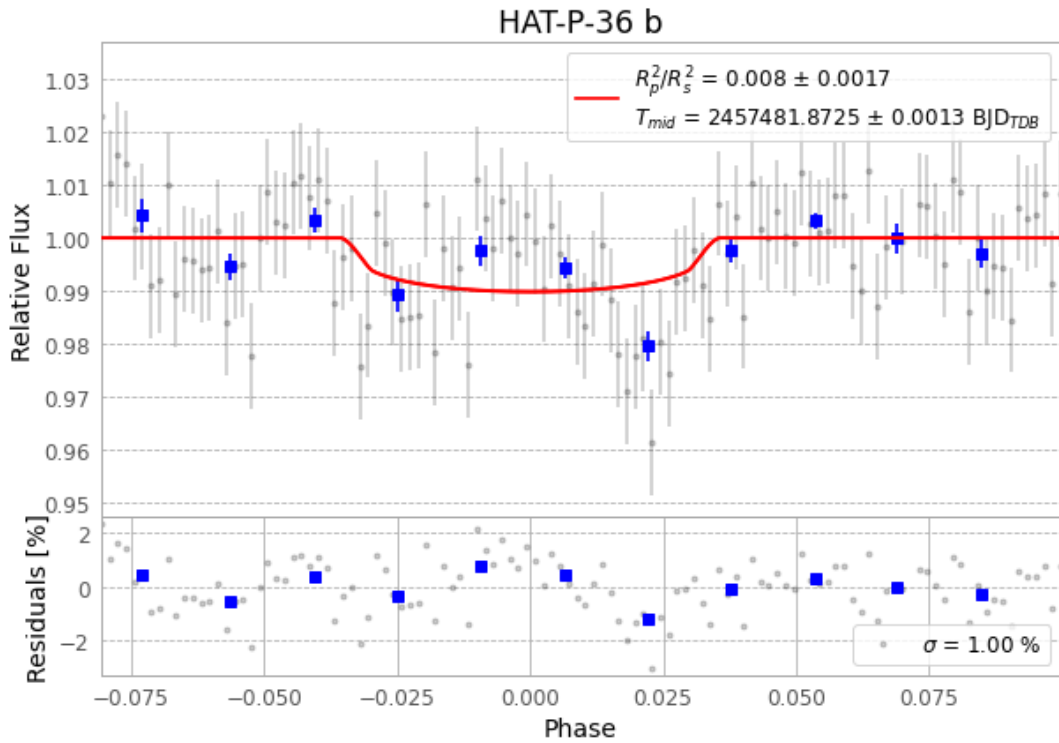


Figure 1 — FinalLightCurve_HAT-P-36 b_2016-04-02.png (SHA-256 9c85eec1b02e9ae3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 10def545e92198f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

116 FITS frames (76 MB), HATP-36160403061512.FITS → HATP-36160403120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-160403.log (69e9a0a02949...), FinalLightCurve_HAT-P-36 b_2016-04-02.png (9c85eec1b02e...), FinalLightCurve_HAT-P-36 b_2016-04-02.csv (51828ff06119...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:25Z.